

# PROJECT\_ANALYSIS.md

## 1. Project Overview

**Project Name:** GarTexHub - B2B Garments & Textile Marketplace

**System Purpose:** An international B2B marketplace connecting buyers (international garment buyers), factories (Bangladeshi textile manufacturers), and buying houses (intermediaries managing multiple factories). The platform features social-media-style feed, real-time messaging, video/audio calls, digital contracts, verification system, and analytics.

**Tech Stack:** - Frontend: React 19 + Vite 8 + Tailwind CSS 4 + Framer Motion - Backend: Node.js + Express 5 - Database: Prisma ORM with PostgreSQL/MySQL support (currently using JSON stores for MVP) - Realtime: WebSocket (ws library) - Authentication: JWT + bcrypt - File Storage: Local filesystem with uploads directory

**Architecture Summary:** - Monolithic Express server with modular route/controller pattern - Frontend SPA with React Router for client-side routing - JSON-based data stores in server/database/ for MVP - WebSocket server integrated in same port for real-time features - Role-based access control (RBAC) with 6 roles: buyer, factory, buying\_house, owner, admin, agent

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## 2. Folder & File Structure (Summarized)

```
meow/
├── server/
│   ├── server.js           # Main Express server + WebSocket
│   ├── controllers/        # 60+ controllers for each domain
│   ├── services/           # Business logic layer
│   ├── routes/             # Express route definitions
│   ├── middleware/         # Auth, validation, error handling
│   ├── utils/              # Utilities (db, logger, validators)
│   ├── database/           # JSON stores (users, requests, products)
│   ├── workers/            # Background job workers
│   ├── realtime/           # WebSocket bus
│   └── uploads/            # File uploads (chat, contracts, feed)
├── src/
│   ├── pages/              # 30+ React page components
│   ├── components/         # Reusable UI components
│   ├── hooks/              # Custom React hooks
│   ├── lib/                # Auth, events, notifications
│   ├── App.jsx             # Main router configuration
│   └── main.jsx            # Entry point
└── prisma/
    └── schema.prisma       # Database schema
```

|                     |                                |
|---------------------|--------------------------------|
| └─ migrations/      | # Migration files              |
| └─ scripts/         | # Build, seeding, CI scripts   |
| └─ tests/           | # Unit, integration, e2e tests |
| └─ electron/        | # Electron desktop app config  |
| └─ package.json     |                                |
| └─ vite.config.js   |                                |
| └─ eslint.config.js |                                |

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### 3. Existing Features (STRICT EXTRACTION)

#### Authentication

- **User Registration** - POST /api/auth/register - Roles: buyer, factory, buying\_house, owner, admin, agent
- **User Login** - POST /api/auth/login - JWT token generation
- **JWT Authentication** - server/middleware/auth.js - Token validation middleware
- **Role-based Access Control** - server/utils/permissions.js - Permission checks per role
- **3-step Onboarding** - Profile image, organization name, category selection - src/pages/auth/OnboardingWizard.jsx

#### UI/UX

- **Combined Feed** - LinkedIn-style feed mixing buyer requests + company products - src/pages/MainFeed.jsx
- **Unique Toggle** - Feed algorithm control (diversified vs personalized) - src/components/feed/FeedControlBar.jsx
- **Floating AI Assistant** - Rule-based guidance on all pages - src/components/FloatingAssistant.jsx
- **Responsive Layout** - Tailwind CSS responsive classes
- **Role-specific Dashboards** - Owner, Agent, Admin panels

#### API

- **User Management** - server/routes/userRoutes.js - Profile CRUD
- **Buyer Requests** - server/routes/requirementRoutes.js - POST/GET/PATCH/DELETE
- **Company Products** - server/routes/productRoutes.js - Upload products
- **Combined Feed** - server/routes/feedRoutes.js - Ranked feed with boost logic
- **Search** - server/routes/searchRoutes.js - Filtered search with quotas
- **Messaging** - server/routes/messageRoutes.js - Real-time chat
- **Conversations** - server/routes/conversationRoutes.js - Thread management
- **Call Sessions** - server/routes/callSessionRoutes.js - Video/audio scheduling

- **Contracts** - server/routes/documentRoutes.js - Draft, sign, artifact management
- **Verification** - server/routes/verificationRoutes.js - Document upload/approval
- **Subscriptions** - server/routes/subscriptionRoutes.js - Free/Premium plans
- **Ratings** - server/routes/ratingsRoutes.js - Submit/view ratings
- **Analytics** - server/routes/analyticsRoutes.js - Dashboard metrics
- **Partner Network** - server/routes/partnerNetworkRoutes.js - Factory connections
- **Member Management** - server/routes/memberRoutes.js - Sub-account management
- **Org Settings** - server/routes/orgRoutes.js - Organization configuration
- **Notifications** - Real-time via WebSocket + server/routes/notificationRoutes.js

## Admin

- **Admin Panel** - src/pages/AdminPanel.jsx - User oversight, verification review
- **Verification Approval** - Manual verification toggle per user
- **Content Moderation** - Report handling, rejection modal
- **Analytics Dashboard** - Platform-wide metrics

## Integrations

- **WebSocket Realtime** - server/realtime/realtimeBus.js - Chat, presence, notifications
- **File Uploads** - Multer for images, PDFs
- **Email (Nodemailer)** - Placeholder for notifications
- **PDF Generation** - PDFKit for contracts
- **OpenSearch** - Search indexing (configured but may not be active)

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## 4. Unfulfilled / Missing Requirements

Based on chat logs analysis:

### Feature: Smart Notifications After Search

- **Requirement:** "Once a user searches for a specific product, the system will send notifications for similar types of products in the future"
- **Evidence:** Chat log - "Smart Notifications: Once a user searches for a specific product, the system will send notifications for similar types of products in the future"
- **Code Status:** Missing
- **Affected Area:** Backend - notification logic + search tracking



## Feature: Full Structured Buyer Request Builder

- **Requirement:** Multi-step form with: Basic Info, Product Specification, Commercial Terms, Compliance
- **Missing Fields:**
  - Quote Deadline (critical)
  - Request Validity/Expiry (critical)
  - Max Suppliers to Contact
  - Preferred Factory Location
  - Factory Size Preference
  - Previous Export Experience
  - Confidentiality Toggle
  - Packaging & Labeling Requirements
  - Shipment Details (Partial Shipment, Mode)
- **Evidence:** Chat log - detailed field requirements for Garments vs Textile
- **Code Status:** Partial - Basic form exists in `src/pages/BuyerRequestManagement.jsx`
- **Affected Area:** Frontend + Backend requirement model

## Feature: Messaging Hierarchy (Verified Priority)

- **Requirement:** Verified users → direct message to inbox; Unverified → message request
- **Evidence:** Chat log - "There verified users will be able to message directly, unverified users will be able to message but they will remain as message requests like Facebook Lite"
- **Code Status:** Partial - Tiered inbox exists, but full enforcement not verified
- **Affected Area:** Backend `messageController.js` + frontend `ChatInterface.jsx`

## Feature: Video Call Recording

- **Requirement:** All calls recorded for security/dispute resolution
- **Evidence:** Chat log - "All calls made through the platform must be recorded for security and record-keeping"
- **Code Status:** Missing - Call interface exists but recording pipeline not found
- **Affected Area:** Backend `callSessionService.js` + frontend `CallInterface.jsx`

## Feature: Secure Documentation & Banking Exchange

- **Requirement:** Platform facilitates exchange of banking info (transaction numbers) for local withdrawals
- **Evidence:** Chat log - "the platform will facilitate the exchange of banking information (such as transaction numbers required for local bank withdrawals)"
- **Code Status:** Partial - Payment proof routes exist but LC workflow incomplete
- **Affected Area:** Backend `paymentProofRoutes.js` + frontend contract flow

## Feature: Digital Contracts (E-sign + PDF Vault)

- **Requirement:** Digital signatures → PDF version to both parties → stored in system
- **Evidence:** Chat log - “users can confirm orders via digital signatures. A PDF version of the contract will be provided to both parties”
- **Code Status:** Partial - Contract draft exists, e-sign workflow incomplete
- **Affected Area:** Backend documentController.js + frontend ContractVault.jsx

## Feature: Content Moderation / Halal Filter

- **Requirement:** Filter content with prohibited musical instruments, obscene content
- **Evidence:** Chat log - “We need a filtering system that will remove any kind of image, photo, etc. with haram musical instruments and obscene content”
- **Code Status:** Missing - No content filter implemented
- **Affected Area:** Backend + frontend feed

## Feature: Anti-Third-Party Communication Sharing

- **Requirement:** Block phone/email/WhatsApp/Telegram/FB/IG/URL sharing
- **Evidence:** Chat log - “we have to be very careful that no third-party communication media can share it through our system”
- **Code Status:** Missing - No anti-contact-sharing implemented
- **Affected Area:** Backend messageController.js

## Feature: Support/Feedback Page

- **Requirement:** Page for bug reports, feature requests, account issues
- **Evidence:** Chat log - “page where people can report website issues and request updates”
- **Code Status:** Partial - SupportReports.jsx exists but may lack full form structure
- **Affected Area:** Frontend SupportReports.jsx

## Feature: Verification Document Matrix

- **Requirement:** Per role (Factory 6 docs, Buying House 5 docs, Buyer 4 docs per region)
- **Evidence:** Chat logs - detailed document lists for Factory, Buying House, Buyer (EU/USA)
- **Code Status:** Partial - Basic verification exists, region-specific docs not implemented
- **Affected Area:** Backend verificationController.js + frontend VerificationPage.jsx

## Feature: Lead Workflow + Enterprise Tools

- **Requirement:** Lead status (New→Closed), assign to agent, internal notes, reminders/follow-ups



- **Evidence:** Chat log - “lead workflow + enterprise tools: lead status (New→Closed), assign to agent, internal notes, reminders/follow-ups”
- **Code Status:** Partial - Lead routes exist, full workflow unclear
- **Affected Area:** Backend leadRoutes.js + frontend lead components

### Feature: Advanced Analytics (5 Key Metrics)

- **Requirements:**
    1. Buyer Request Match Rate
    2. Lead → Deal Conversion Rate
    3. Factory Response Speed
    4. Buyer Demand Trend
    5. Trusted Deal Score
  - **Evidence:** Chat log - “Advanced Analytics: 5 key metrics”
  - **Code Status:** Partial - Basic analytics exists, custom metrics not fully implemented
  - **Affected Area:** Backend analyticsService.js + frontend Insights.jsx
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## 5. Frontend Problems (STRICT)

### Issue: Onboarding Shows on Every Page Load

- **Description:** The “Unique” button and other feed controls appear on all pages during account creation, causing confusion
- **File:** src/pages/auth/OnboardingWizard.jsx
- **Why Problem:** Chat log: “general just only account creation time why unique button and other things showing! This is a serious problem”

### Issue: Profile Page Information Leakage

- **Description:** Clicking “View Details” on buyer profile shows all information to outside users
- **File:** src/pages/BuyerProfile.jsx
- **Why Problem:** Chat log: “the work outside the company can get all the information - that doesn’t seem right”

### Issue: Navigation Links Too Prominent

- **Description:** All links visible at once, showing unnecessary buttons to users
- **File:** src/App.jsx, navigation components
- **Why Problem:** Chat log: “all links are being seen at once a little problem is happening, some unnecessary buttons are there”

### Issue: Missing Industry Field on Profiles

- **Description:** All account types should show industry category (Garments, Textile, Buying House, Buyer) prominently

- **File:** src/pages/BuyerProfile.jsx, src/pages/FactoryProfile.jsx, src/pages/BuyingHouseProfile.jsx
- **Why Problem:** Chat log: "Industry: Garments, Organization: Direct Buyer, Rating: 4.6/5 - this method is the most correct" - not consistently applied

#### Issue: Rating/Review Deletion by Wrong User

- **Description:** Users can delete their own reviews, but should only be deletable by the reviewer
- **File:** src/pages/RatingFeedback.jsx
- **Why Problem:** Chat log: "reviews page and rating page must be there - reviews to be made so that the person whose profile it is can also delete if they want, but only the person who gave the review can delete or edit"

#### Issue: Feedback/Support Form Incomplete

- **Description:** Support page lacks proper form structure with categories, screenshot upload, priority levels
- **File:** src/pages/SupportReports.jsx
- **Why Problem:** Chat log details required fields: Subject, Category, Description, Screenshot upload, Page URL, Priority level, Contact email

#### Issue: Missing Deadline Option on Buyer Requests

- **Description:** Buyer request form lacks deadline field for quote submissions
- **File:** src/pages/BuyerRequestManagement.jsx
- **Why Problem:** Chat log: "deadline option, this should be optional for users to use"

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## 6. Bugs & Technical Issues

#### Issue: Missing Input Validation in Login API

- **Location:** server/controllers/authController.js
- **Severity:** High
- **Impact:** Allows malformed requests without proper email/password validation
- **Fix:** Add schema validation (e.g., Joi/Zod) for login endpoint

#### Issue: JSON Store Race Conditions

- **Location:** server/utils/jsonStore.js
- **Severity:** High
- **Impact:** Concurrent writes to JSON files can cause data loss or corruption
- **Fix:** Implement file locking or migrate to proper database

#### Issue: No Rate Limiting on Search API

- **Location:** server/routes/searchRoutes.js

- **Severity:** Medium
- **Impact:** Daily search quota can be bypassed
- **Fix:** Add rate limiting middleware

#### Issue: WebSocket Memory Leak - Socket Cleanup

- **Location:** server/server.js lines ~200+
- **Severity:** Medium
- **Impact:** Long-running connections may not be cleaned up properly
- **Fix:** Add proper socket cleanup in connection close handler

#### Issue: JWT Secret in Code

- **Location:** server/server.js line 196
- **Severity:** Critical
- **Impact:** Hardcoded JWT secret "mvp-dev-secret" - security risk
- **Fix:** Move to environment variable with validation

#### Issue: Missing Error Handling in File Uploads

- **Location:** server/controllers/documentController.js
- **Severity:** Medium
- **Impact:** Upload failures can crash the server
- **Fix:** Add try-catch blocks and proper error responses

#### Issue: CORS Wildcard Configuration

- **Location:** server/server.js line 102
- **Severity:** Medium
- **Impact:** Allows any origin to access API
- **Fix:** Restrict to specific origins in production

#### Issue: No Input Sanitization on HTML Content

- **Location:** server/services/chatbotService.js, server/controllers/feedPostController.js
- **Severity:** High
- **Impact:** XSS vulnerabilities in markdown rendering
- **Fix:** Sanitize user input before storage/rendering

#### Issue: Duplicate Code in Controller Logic

- **Location:** Multiple controllers
- **Severity:** Low
- **Impact:** Maintainability issues
- **Fix:** Extract common logic to services



## Issue: Missing Database Indexes

- **Location:** prisma/schema.prisma
  - **Severity:** Medium
  - **Impact:** Slow queries on large datasets
  - **Fix:** Add indexes on frequently queried fields (userId, role, createdAt)
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## 7. Code Quality & Architecture Review

### Modularity: Good

- Controllers separated by domain
- Services layer for business logic
- Middleware for cross-cutting concerns

### Reusability: Good

- Shared utilities in server/utils/
- Frontend hooks for common state management

### Separation of Concerns: Needs Improvement

- Server.js is 819 lines - too large, should extract WebSocket logic
- Controllers do too much - validation + business logic

### Scalability: Concerns

- JSON file stores won't scale - needs real database
- No caching layer (Redis not integrated despite being in tech stack)
- WebSocket not clustered - single instance limitation

### Concrete Improvements Needed:

1. **Extract WebSocket logic** to server/realtime/server.js separate from main server
  2. **Add Redis** for session/caching (in tech stack but not used)
  3. **Implement proper database** - Prisma is set up but JSON still used
  4. **Add API versioning** - /api/v1/ prefix for future compatibility
  5. **Centralize error handling** - current scattered try-catch blocks
  6. **Add request validation middleware** - central schema validation
  7. **Extract constants** - hardcoded values should be in config
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## 8. Actionable Fix Plan

### Phase 1: Critical Fixes (Immediate)

1. **Fix JWT secret hardcoding** - Move to env with validation

2. **Fix XSS vulnerabilities** - Sanitize all user inputs
3. **Fix CORS configuration** - Restrict origins in production
4. **Fix onboarding UI bug** - Prevent feed controls showing on signup

## Phase 2: Feature Completion (High Priority)

1. **Complete Buyer Request Builder** - Add all missing fields (deadline, supplier preferences, etc.)
2. **Implement messaging hierarchy** - Full verified/unverified message routing
3. **Complete verification documents** - Region-specific (EU/USA) document sets
4. **Implement payment proof workflow** - Bank transfer + LC states
5. **Complete contract e-sign flow** - Digital signatures + PDF generation

## Phase 3: UI/UX Improvements

1. **Fix profile information leakage** - Proper visibility controls
2. **Add support form structure** - Full feedback form with categories/uploads
3. **Fix rating review permissions** - Only reviewer can edit/delete
4. **Apply consistent industry labels** - All profile types show industry category
5. **Add "Unique" toggle on feed** - Algorithm control feature

## Phase 4: Missing Systems (Medium Priority)

1. **Implement video call recording** - Storage + playback
2. **Add content moderation** - Halal/prohibited content filter
3. **Implement anti-contact-sharing** - Block third-party communication
4. **Add smart notifications** - Post-search product alerts

## Phase 5: Refactoring

1. **Migrate to proper database** - Replace JSON stores with PostgreSQL
2. **Add Redis caching** - For sessions and frequently accessed data
3. **Extract WebSocket server** - Separate file for clarity
4. **Add centralized validation** - Zod schemas for all endpoints

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## Summary

The GarTexHub project has implemented a solid foundation for a B2B marketplace with authentication, feed, messaging, and basic admin functionality. However, several key features from the client's requirements are incomplete or missing, particularly around the buyer request workflow, payment/contract processes, content moderation, and advanced analytics. The codebase would benefit from security hardening (JWT, XSS, CORS), proper database migration, and UI fixes as identified in the issues section.